DO NOW — RECALL

TCP/IP stack, protocols and DNS

Name: _____ Class: _____ Date: _____

1. Which device typically forwards packets between different IP networks?

2. What does an edge represent in a network graph?

3. How many bits make one byte?

4. A sender and receiver are on different networks. Can a switch alone normally route IP packets between them?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Router.

2. Model answer

A connection between two vertices.

3. Model answer

Eight.

4. Model answer

No. A router (or routing-capable layer-3 switch) is needed to route between IP networks.

Vocabulary — know these terms

application layer — Protocols offering services to applications, including DNS and HTTP.

transport layer — End-to-end transport protocols such as TCP and UDP.

internet layer — Network addressing and forwarding using IP.

link layer — Local transfer of frames over media such as Ethernet or Wi-Fi.

DNS — Domain Name System; supports name-to-address resolution.

NEW LEARNING

Read, annotate and explain

Start with a hostname

The browser uses DNS (an application-layer protocol) to resolve school.example to an IP address. A cached result may avoid further queries.

Understand four layers

Application supports services such as HTTP(S), DNS and SMTP; transport includes TCP and UDP; internet uses IP; link transfers frames on the local connection.

Build an HTTPS connection

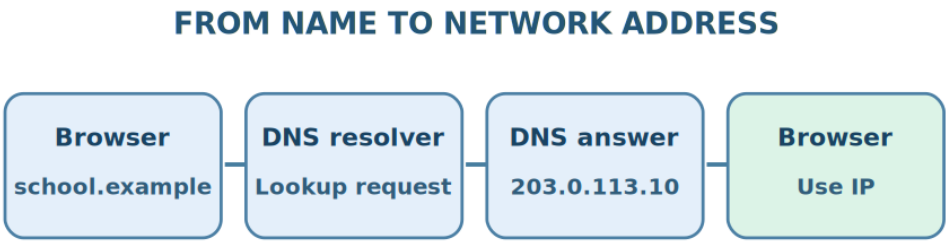
A browser normally establishes TCP to port 443, negotiates TLS, then exchanges encrypted HTTP requests and responses. DNS lookup happens first if the IP is unknown.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

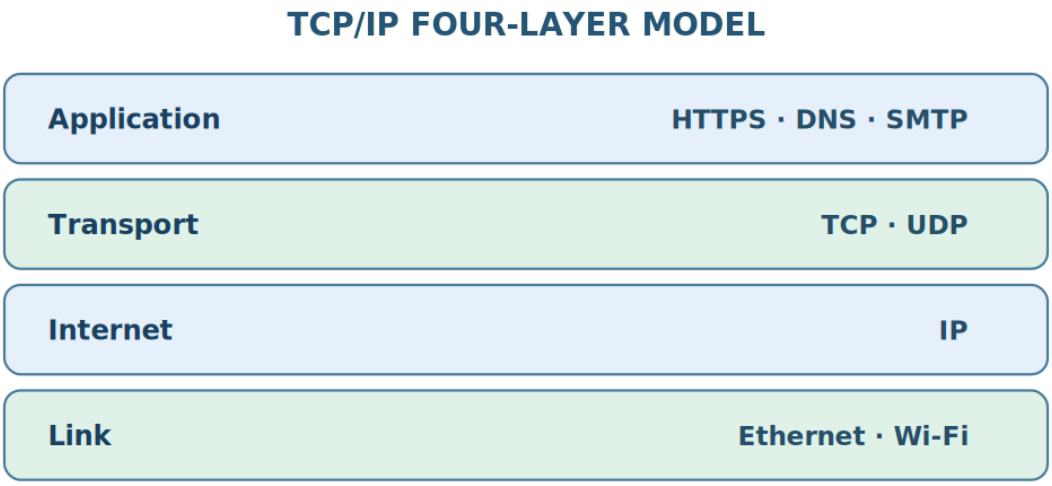
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



FROM NAME TO NETWORK ADDRESS. Browser: school.example; DNS resolver: Lookup request; DNS answer: 203.0.113.10; Browser: Use IP

New Learning

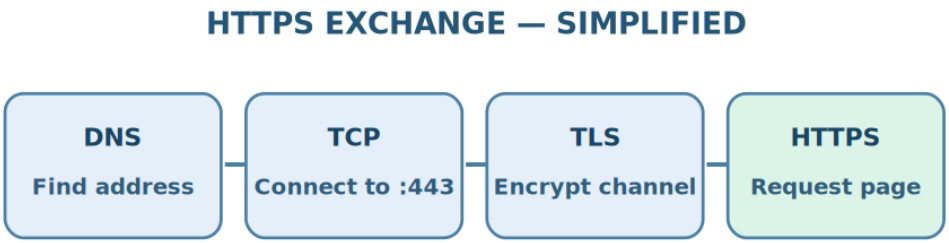


TCP/IP FOUR-LAYER MODEL. Application includes HTTPS · DNS · SMTP; Transport includes TCP · UDP; Internet includes IP; Link includes Ethernet · Wi-Fi

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



HTTPS EXCHANGE — SIMPLIFIED. DNS: Find address; TCP: Connect to :443; TLS: Encrypt channel; HTTPS: Request page

I DO — WORKED EXAMPLE

Follow the worked method

A user opens https://library.example. Explain what DNS, TCP and HTTPS each do.

Name resolution

The resolver returns the IP address for library.example (or uses a cached answer).

Transport connection

TCP establishes a connection to the server, typically destination port 443.

Application exchange

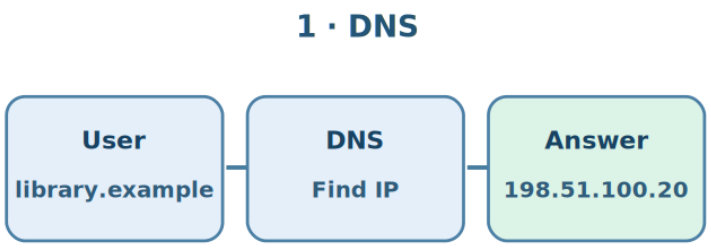
TLS secures the connection; HTTPS carries the page request and response.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A user opens https://library.example. Explain what DNS, TCP and HTTPS each do.

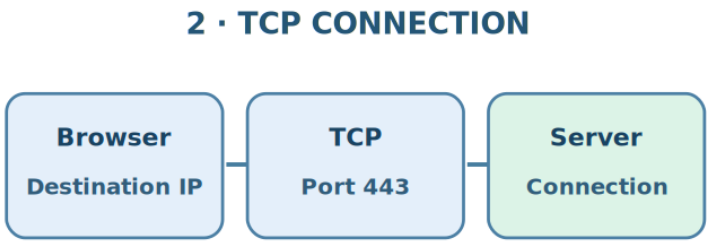
The resolver returns the IP address for library.example (or uses a cached answer).



1 · DNS. User: library.example; DNS: Find IP; Answer: 198.51.100.20

A user opens https://library.example. Explain what DNS, TCP and HTTPS each do.

The resolver returns the IP address for library.example (or uses a cached answer).



2 · TCP CONNECTION. Browser: Destination IP; TCP: Port 443; Server: Connection

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A user opens https://library.example. Explain what DNS, TCP and HTTPS each do.

The resolver returns the IP address for library.example (or uses a cached answer).



3 · ENCRYPTED REQUEST. TLS: Secure; HTTPS: GET /; Server: Response

WE DO — GUIDED PRACTICE

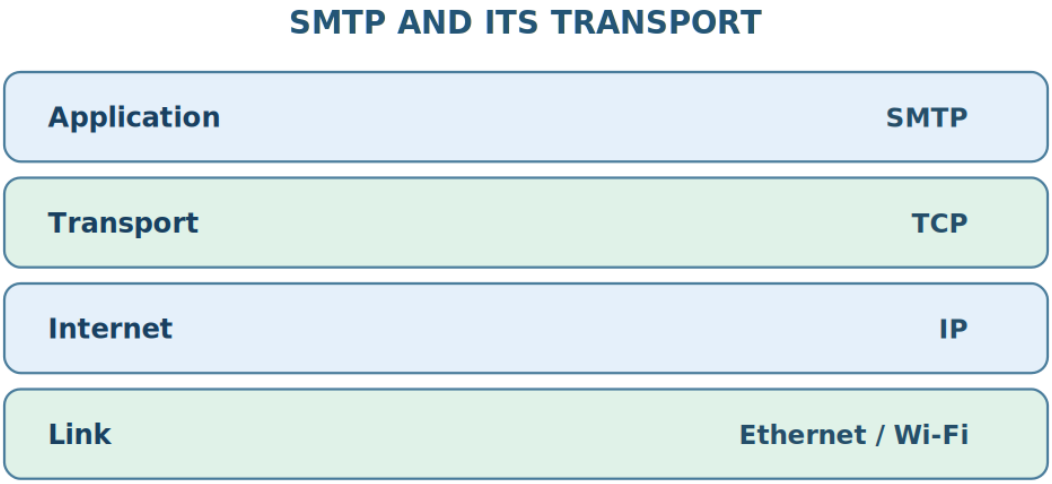
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil sends email using SMTP. Place SMTP, TCP and IP into the correct TCP/IP layers.

SMTP is an application-layer protocol for transferring outgoing email.



SMTP AND ITS TRANSPORT. Application includes SMTP; Transport includes TCP; Internet includes IP; Link includes Ethernet / Wi-Fi

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Identify the application SMTP is an application-layer protocol for transferring outgoing email.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Which TCP/IP layer contains DNS, and what does DNS do? [2]

Question 2

Place TCP, IP, and HTTP into the correct layers and explain one role of each. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A pupil says DNS encrypts webpage traffic and TCP chooses the Wi-Fi channel. Correct both errors, identifying the relevant protocols/layers. [4]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

DNS and HTTPS are both application-layer protocols in the TCP/IP model.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

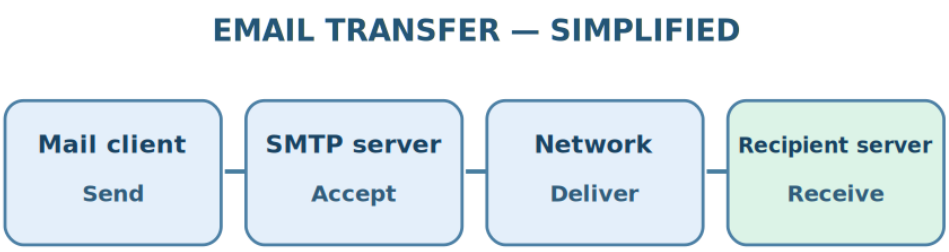
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil sends email using SMTP. Place SMTP, TCP and IP into the correct TCP/IP layers.

SMTP is an application-layer protocol for transferring outgoing email.



EMAIL TRANSFER — SIMPLIFIED. Mail client: Send; SMTP server: Accept; Network: Deliver; Recipient server: Receive

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Identify the other layers

TCP provides reliable transport for SMTP; IP handles inter-network addressing and routing; link delivers frames on each local link.

IMPROVE: Explain one correction to your own reasoning.